N S Vaishnavi

Aspiring Bioinformatician

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PROFESSIONAL SUMMARY

Bioinformatics graduate with a strong foundation in **Python, R, and statistical modeling**. Experienced in **data triage, dataset validation, and building predictive models**. Proven ability to handle large-scale datasets and technical documentation within regulated environments, with a focus on data integrity and process automation.

PROFESSIONAL EXPERIENCE

Reliance Life Sciences Pvt. Ltd, *Graduate Engineer Trainee in Manufacturing* — 11/2024 – 11/2025, Navi Mumbai, India

- Gained hands-on experience in **Downstream bioprocessing techniques** adhering to **cGMP** and **SOPs.**
- Supported the implementation and optimization of **Electronic Batch Manufacturing Records (eBMR)**, ensuring data integrity.
- Maintained **GMP-compliant documentation** adhering to **ALCOA++ principles**, focusing on the accuracy and reliability of manufacturing data.
- Partnered with technical teams to resolve issues and optimise data-driven workflows.

BiomarkIQ, *Biomarker Discovery and Analysis - Intern* — 09/2024 – 11/2024, Remote

- **Data Transformation:** Performed data triage and quality checks on clinical datasets using Python and R to ensure data integrity for research.
- **Reporting & Visualization:** Prepared structured summaries and **data visualizations** to support biomarker discovery, aligning with specific research inclusion/exclusion criteria.
- **Workflow Optimization:** Improved data-handling workflows to increase efficiency in processing large research datasets.

EDUCATION

B.Tech - Bioinformatics, SASTRA *Deemed to be University* — 2020 – 2024
CGPA : 7.35 — Thanjavur

12th grade, *Chellammal Matriculation Higher Secondary School* — 2018 – 2020
Marks: 515/600 (85.83%) — Tiruchirapalli

SKILLS

Programming & Databases: Python (Pandas, NumPy), R, SQL (Basics), GitHub. | **Data Visualization:** Matplotlib, Seaborn, Excel (Advanced), PowerPoint. | **Statistics:** Logistic Regression, Hypothesis Testing, Dataset Validation, Data Cleaning. | **Tools:** Microsoft Excel, PowerPoint, Schrödinger Maestro, NCBI Databases.

PROJECTS

Design and Evaluation of Novel Inhibitors to Combat Antimicrobial Drug Resistance, — 01/2024 – 05/2024
In-silico Drug Design | DNA Isolation | PCR

- Performed structure-based virtual screening targeting enzymes in the menaquinone pathway.
- Assessed ADME profiles and shortlisted hit molecules.
- Conducted MD simulations to study protein–ligand interactions.
- Tools: Schrödinger Maestro.

Identifying genetic signatures for early detection of diabetic complications, — 07/2023 – 11/2023
Gene Expression Data Analysis | R Programming | Logistic Regression | Pathway Analysis

- **Statistical Modeling:** Developed a **Logistic Regression model** to assess biomarker predictive accuracy.
- **Data Extraction:** Utilized R to mine and analyze **Gene Expression (GEO) datasets**, performing differential expression and pathway analysis.
- **Insight Generation:** Conducted functional enrichment and pathway-level interpretation to turn raw data into actionable biological insights.

The Seattle Structural Genomics Centre for Infectious Disease (SSGCID) Approach for Structure-Guided Drug Discovery with Brain-Eating Trophozoites, — 05/2023
Literature Review| Structural Genomics | Target & Pathogenesis Analysis

- Conducted a comprehensive analysis of the SSGCID's pipeline for developing novel therapeutics against *Balamuthia mandrillaris*.
- Evaluated high-throughput screening and structural biology studies to identify critical protein targets and map their roles in pathogen virulence and metabolic pathways.
- Investigated the identification of novel candidates and the repurposing of existing drugs through structure-based workflows.

CERTIFICATIONS

Python for Data Science, AI and Development: *issued by Coursera* | **Python Data Structure and Algorithms:** *issued by LinkedIn Learning*

EXTRA CURRICULAR ACTIVITIES

- Executive head at E-cell SASTRA, the entrepreneurship club of SASTRA University
- A key member of Gaanavarshini, the official Carnatic music club of SASTRA University.
- A member of KS Upahaar, the charitable and fundraising club of SASTRA University.